

Calculus 2

8 Lecture 8

Analytic Functions

Definition 8.1. Let $n \in \mathbb{N} \cup \{0\}$ and let $I \subseteq \mathbb{R}$ be an open interval. Then, a function $f : I \rightarrow \mathbb{R}$ is said to be

- a) *n-differentiable* on a set $E \subseteq I$ if the derivatives

$$f^0 = f, f', f'', \dots, f^{(n)}$$

exist at every point of E ;

- b) of *differentiable class* $C^n(E)$ if the derivatives

$$f^0 = f, f', f'', \dots, f^{(n)}$$

exist and are continuous at every point of E ;

- c) *smooth* (or *infinitely differentiable*) on E , denoted $f \in C^\infty(E)$, if it has derivatives of all orders at every point of E .

It ought to be clear that (see Calculus-1)

$$C^n(E) \subset C^{n-1}(E), \quad C^\infty(E) = \bigcap_{n=0}^{\infty} C^n(E).$$

Definition 8.2. Let $I \subseteq \mathbb{R}$ be an open interval and let a function $f : I \rightarrow \mathbb{R}$ be smooth at a point $\lambda \in I$ (i.e. $f \in C^\infty(\{\lambda\})$). Then, the *Taylor series* of the function $f(x)$ at the point λ (denoted $T_{f,\lambda}(x)$) is the power series of the form

$$T_{f,\lambda}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k = f(\lambda) + f'(\lambda)(x - \lambda) + \frac{f''(\lambda)}{2!} (x - \lambda)^2 + \dots \quad (1)$$

Coefficients $\frac{f^{(k)}(\lambda)}{k!}$ are called the *Taylor coefficients*.

Definition 8.3. Let $I \subseteq \mathbb{R}$ be an open interval. Then a function $f : I \rightarrow \mathbb{R}$ is said to be *analytic* at a point $\lambda \in I$, denoted $f \in C^\omega(\{\lambda\})$, if there is a sequence $\{a_k\}_{k=0}^{\infty} \subset \mathbb{R}$ and $\delta > 0$ such that

$$f(x) = \sum_{k=0}^{\infty} a_k (x - \lambda)^k, \quad \forall x \in (\lambda - \delta, \lambda + \delta).$$

That is, a function f is analytic at a point $\lambda \in I$ if it can be given by a convergent power series centered at λ in some neighbourhood of λ .

A function f is *analytic on an open interval* $J \subseteq I$ if it is analytic at every point $\lambda \in J$.

Claim 8.1 (Any Analytic Function is Smooth). Let $I \subseteq \mathbb{R}$ be an open interval. If a function $f : I \rightarrow \mathbb{R}$ is analytic at a point $\lambda \in I$ (i.e. $f \in C^\omega(\{\lambda\})$), then:

1. f is smooth at λ (i.e. $f \in C^\infty(\{\lambda\})$);
- 2.

$$f(x) = T_{f,\lambda}(x), \quad \forall x \in (\lambda - \delta, \lambda + \delta),$$

for some $\delta > 0$, where $T_{f,\lambda}(x)$ is the Taylor series of the function $f(x)$ at the point λ (see [Def 8.2](#)).

Proof. 1) Due to [Def 8.3](#), we have

$$f(x) = \sum_{n=0}^{\infty} a_n(x - \lambda)^n, \quad \forall x \in (\lambda - \delta, \lambda + \delta), \quad (1)$$

for some $\delta > 0$. Therefore, due to [Def 7.2](#), the series $\sum_{n=0}^{\infty} a_n(x - \lambda)^n$ has a positive radius of convergence $R \geq \delta > 0$. Thus, due to [Claim 7.6](#), [Equality \(1\)](#) implies that, for every $k \in \mathbb{N} \cup \{0\}$ and every $x \in (\lambda - \delta, \lambda + \delta)$, we have

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n (x - \lambda)^{n-k}.$$

If we substitute $x = \lambda$, then

$$f^{(k)}(\lambda) = k! a_k, \quad a_k = \frac{f^{(k)}(\lambda)}{k!}, \quad \forall k \in \mathbb{N} \cup \{0\}. \quad (2)$$

Thus, due to [Def 8.1.c](#)), f is smooth on $E = \{\lambda\}$.

2) Note that, $\forall x \in (\lambda - \delta, \lambda + \delta)$, we have

$$f(x) \stackrel{1}{=} \sum_{k=0}^{\infty} a_k (x - \lambda)^k \stackrel{2}{=} \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k \stackrel{3}{=} T_{f, \lambda}(x).$$

Here 1 holds due to [Equality \(1\)](#), 2 holds due to [Equality \(2\)](#), and 3 holds due to [Def 8.2](#) and since $f \in C^\infty(\{\lambda\})$ due to [Item 1](#). $\square$

Claim 8.2 (Example of non-analytic smooth function). Let $\tilde{\tau} : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$\tilde{\tau}(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Then:

1.

$$\tilde{\tau}^{(n)}(x) = \begin{cases} \frac{P_n(x)}{x^{2n}} \tilde{\tau}(x), & x > 0, \\ 0, & x \leq 0, \end{cases} \quad \forall n \in \mathbb{N}, \quad (3)$$

where $P_n(x)$ is a polynomial of degree $n - 1$ given recursively by $P_1(x) = 1$ and

$$P_{n+1}(x) = x^2 P'_n(x) - (2nx - 1)P_n(x), \quad \forall n \geq 1;$$

2. $\tilde{\tau} \in C^\infty(\mathbb{R})$ and $\tilde{\tau}^{(n)}(0) = 0, \forall n \in \mathbb{N} \cup \{0\}$;

3. $\tilde{\tau}$ is not analytic at $\lambda = 0$.

Proof. 1) and 2) One can prove by math. induction on $n \in \mathbb{N}$. We leave it as an exercise for the reader.

3) Let us assume that $\tilde{\tau}(x)$ is analytic at $\lambda = 0$. Then, for some $\delta > 0$ and $\forall x \in (0 - \delta, 0 + \delta) = (-\delta, \delta)$ we have

$$\tilde{\tau}(x) = T_{\tilde{\tau}, 0}(x) = \sum_{k=0}^{\infty} \frac{\tilde{\tau}^{(k)}(0)}{k!} (x - 0)^k = \sum_{k=0}^{\infty} 0 \cdot x^k = 0. \quad (4)$$

The first equality holds due to [Item 2\)](#) of [Claim 8.1](#) (since we assumed that $\tilde{\tau} \in C^\omega(\{0\})$), the second due to [Def 8.2](#) and since $\tilde{\tau} \in C^\infty(\{0\}) \subset C^\infty(\mathbb{R})$, and the third since $\tilde{\tau}^{(k)}(0) = 0, \forall k \in \mathbb{N} \cup \{0\}$.

[Equality \(4\)](#) contradicts the definition of $\tilde{\tau}$ (indeed, $\forall x > 0$ we have $\tilde{\tau}(x) = e^{-1/x} > 0$). Thus, our assumption is false and $\tilde{\tau}(x)$ is not analytic at $\lambda = 0$. $\square$

Later we will need the following well-known fact.

Claim 8.3 (The Taylor Formula). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$ be an $(n+1)$ -differentiable on I function (see [Def 8.1.a](#)), and let $\lambda \in I$. Then, for each $x \in I$, there is a point β located between λ and x such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k + R_n(x), \quad (5)$$

where the remainder term $R_n(x)$ can be written as

$$\bullet \quad R_n(x) = \frac{f^{(n+1)}(\beta)}{(n+1)!} (x - \lambda)^{n+1} \quad [\text{the Lagrange form of } R_n(x)]$$

or

$$\bullet \quad R_n(x) = \frac{f^{(n+1)}(\beta)}{n!} (x - \beta)^n (x - \lambda) \quad [\text{the Cauchy form of } R_n(x)].$$

Proof. See Calculus-1. □

Claim 8.4 (Criterion for Analyticity). Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$ be a smooth function on I (i.e. $f \in C^\infty(I)$). Then, the following two conditions are equivalent:

1. f is analytic at $\lambda \in I$ (i.e. $f \in C^\omega(\{\lambda\})$);
2. there exist constants $L > d > 0$ and $C > 0$ such that

$$(\lambda - d, \lambda + d) \subset I \quad \text{and} \quad |f^{(k)}(x)| \leq C \cdot \frac{k!}{L^k}$$

for all $k \in \mathbb{N} \cup \{0\}$ and all $x \in (\lambda - d, \lambda + d)$.

Proof. 2) $\Rightarrow$ 1) Due to [Def 8.3](#), it suffices to prove that

$$\exists \delta > 0 \quad \forall x \in (\lambda - \delta, \lambda + \delta) \quad \rightarrow \quad f(x) = T_{f,\lambda}(x), \quad (6)$$

where $T_{f,\lambda}(x)$ is the Taylor series of f at λ (see [Def 8.2](#)).

Since $f \in C^\infty(I)$, conditions of [Claim 8.3](#) are satisfied for $\forall n \in \mathbb{N}$ and, due to [Formula \(5\)](#), we have

$$f(x) = T_{f,\lambda}^n(x) + R_n(x), \quad \forall n \in \mathbb{N} \cup \{0\}, \quad \forall x \in I, \quad (7)$$

where

$$T_{f,\lambda}^n(x) = \sum_{k=0}^n \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k$$

is the n -th partial sum of $T_{f,\lambda}(x)$ and

$$R_n(x) = \frac{f^{(n+1)}(\beta)}{(n+1)!} (x - \lambda)^{n+1}$$

for some β between λ and x .

Since, due to [Equality \(7\)](#), for every $x \in I$ we have

$$\lim_{n \rightarrow \infty} R_n(x) = \lim_{n \rightarrow \infty} (f(x) - T_{f,\lambda}^n(x)) = f(x) - \lim_{n \rightarrow \infty} T_{f,\lambda}^n(x),$$

[Condition \(6\)](#) can be written as

$$\exists \delta > 0 \quad \forall x \in (\lambda - \delta, \lambda + \delta) \quad \rightarrow \quad \lim_{n \rightarrow \infty} R_n(x) = 0. \quad (8)$$

Let $\delta = d$, where $d > 0$ is from [Item 2](#)), then, $\forall x \in (\lambda - d, \lambda + d)$, we have

$$|R_n(x)| \leq C \cdot \frac{(n+1)!}{L^{n+1}} \cdot \frac{1}{(n+1)!} \cdot d^{n+1} = C \left(\frac{d}{L}\right)^{n+1}. \quad (9)$$

Indeed, the first inequality holds due to the definition of $R_n(x)$, since β is between λ and x , hence $\beta \in (\lambda - d, \lambda + d)$, and since $|x - \lambda| < d$. Since $L > d > 0$, we have

$$\lim_{n \rightarrow \infty} C \left(\frac{d}{L}\right)^{n+1} = 0.$$

Therefore, [Inequality \(9\)](#) implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for every $x \in (\lambda - d, \lambda + d)$. Thus [Condition \(8\)](#) (and [\(6\)](#), since they are equivalent) is proved.

1) $\Rightarrow$ 2) First note that if we differentiate $n \in \mathbb{N} \cup \{0\}$ times term-by-term the well-known equality

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad \forall x \in (-1, 1),$$

due to [Claim 7.6](#), we will receive

$$\sum_{k=n}^{\infty} k^n x^{k-n} = \frac{n!}{(1-x)^{n+1}}, \quad \forall x \in (-1, 1), \quad (10)$$

where

$$k^n = \begin{cases} k(k-1) \cdots (k-n+1), & n \in \mathbb{N}, \\ 1, & n = 0. \end{cases}$$

Due to [Item 2](#)) of [Claim 8.1](#), $f \in C^\omega(\{\lambda\})$ implies that

$$f(x) = T_{f,\lambda}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x-\lambda)^k, \quad \forall x \in (\lambda - \delta, \lambda + \delta) \subset I \quad (11)$$

for some $\delta > 0$. Note that, due to [Def 7.2](#) and [Equality \(11\)](#), the radius of convergence of the series $T_{f,\lambda}(x)$ is at least $\delta > 0$. Let $p = \frac{3}{4}\delta > 0$, then, due to [Claim 7.9](#),

$$\exists \widehat{C} > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad \left| \frac{f^{(k)}(\lambda)}{k!} \right| \leq \frac{\widehat{C}}{p^k}. \quad (12)$$

Let $d = \frac{1}{4}\delta > 0$, then

$$(\lambda - d, \lambda + d) \subset (\lambda - \delta, \lambda + \delta) \subset I.$$

If we differentiate n times term-by-term [Equality \(11\)](#), due to [Claim 7.6](#), we will receive

$$f^{(n)}(x) = \sum_{k=n}^{\infty} k^n \frac{f^{(k)}(\lambda)}{k!} (x-\lambda)^{k-n}, \quad \forall x \in (\lambda - \delta, \lambda + \delta), \quad (13)$$

where coefficients k^n are defined as in [\(10\)](#).

Let $y \in (\lambda - d, \lambda + d)$.

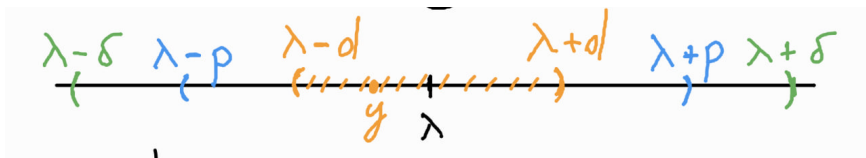


Figure 8.1

Then, for $\forall n \in \mathbb{N} \cup \{0\}$, we have

$$\begin{aligned}
|f^{(n)}(y)| &\leq \sum_{k=n}^{\infty} k^n \left| \frac{f^{(k)}(\lambda)}{k!} \right| |y - \lambda|^{k-n} \\
&\leq \sum_{k=n}^{\infty} k^n \frac{\widehat{C}}{p^k} |y - \lambda|^{k-n} \\
&= \frac{\widehat{C}}{p^n} \sum_{k=n}^{\infty} k^n \left(\frac{|y - \lambda|}{p} \right)^{k-n} \\
&= \frac{\widehat{C}}{p^n} \cdot \frac{n!}{\left(1 - \frac{|y - \lambda|}{p}\right)^{n+1}} \\
&= \frac{p\widehat{C}n!}{(p - |y - \lambda|)^{n+1}} \\
&\leq \frac{p\widehat{C}n!}{(\frac{1}{2}\delta)^{n+1}} = \frac{2p\widehat{C}}{\delta} \cdot \frac{n!}{(\frac{1}{2}\delta)^n} = C \cdot \frac{n!}{L^n},
\end{aligned}$$

where

$$C = \frac{2p\widehat{C}}{\delta} = \frac{2 \cdot \frac{3}{4}\delta\widehat{C}}{\delta} = \frac{3}{2}\widehat{C} > 0, \quad L = \frac{1}{2}\delta > 0.$$

Here we used [Formula \(10\)](#), $\frac{|y - \lambda|}{p} < \frac{d}{p} = \frac{1}{3} < 1$, and, since $p = \frac{3}{4}\delta$ and $|y - \lambda| < d = \frac{1}{4}\delta$, we have $p - |y - \lambda| > \frac{1}{2}\delta$. Since $L = \frac{1}{2}\delta > \frac{1}{4}\delta = d > 0$, the statement is proved. $\square$

Claim 8.5. 1. the functions e^x , $\sin x$, and $\cos x$ are analytic on $\mathbb{R}$;

2. $\ln x$ is analytic on $(0, +\infty)$.

Proof. 2) Due to [Claim 8.4](#), it suffices to show that for every $\lambda \in (0, +\infty)$ there are constants $L > d > 0$ and $C > 0$ such that

$$|((\ln x)^{(k)})| \leq C \cdot \frac{k!}{L^k}, \quad \forall x \in (\lambda - d, \lambda + d) \subset (0, +\infty).$$

Let $d = \frac{1}{4}\lambda$ and $L = \frac{1}{2}\lambda$. Then

$$(\lambda - d, \lambda + d) = \left(\frac{3}{4}\lambda, \frac{5}{4}\lambda\right) \subset (0, +\infty),$$

and, for every $k \in \mathbb{N}$ and every $x \in (\frac{3}{4}\lambda, \frac{5}{4}\lambda)$, we have

$$|((\ln x)^{(k)})| = \frac{(k-1)!}{x^k} \leq (k-1)! \left(\frac{4}{3\lambda}\right)^k \leq \frac{k!}{L^k}.$$

For $k = 0$, the function $\ln x$ is bounded on the closed interval $[\frac{3}{4}\lambda, \frac{5}{4}\lambda]$. Hence, choosing

$$C = \max \left\{ 1, \max_{x \in [\frac{3}{4}\lambda, \frac{5}{4}\lambda]} |\ln x| \right\},$$

we obtain

$$|((\ln x)^{(k)})| \leq C \cdot \frac{k!}{L^k} \quad \forall k \in \mathbb{N} \cup \{0\}, \quad \forall x \in (\lambda - d, \lambda + d),$$

where $L = \frac{1}{2}\lambda > d = \frac{1}{4}\lambda > 0$.

1) We leave it as exercise (similar to [Item 2](#)). $\square$

We will use the following claim to construct an important example of a smooth non-analytic function.

Claim 8.6 (The Borel Lemma). For every sequence $\{\alpha_n\}_{n=0}^\infty$ there exists a smooth function f on $\mathbb{R}$ such that

$$f^{(n)}(0) = \alpha_n, \quad \forall n \in \mathbb{N} \cup \{0\}.$$

Proof. For $\forall x \in \mathbb{R}$ let us consider the following functions

$$\tilde{\tau}(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad g(x) := \frac{\tilde{\tau}(x)}{\tilde{\tau}(x) + \tilde{\tau}(1-x)}, \quad h(x) := g(2+x)g(2-x).$$

$\tilde{\tau} \in C^\infty(\mathbb{R})$ due to Item 2 of [Claim 8.2](#).

$g \in C^\infty(\mathbb{R})$ since $\tilde{\tau}(x) + \tilde{\tau}(1-x) > 0, \forall x \in \mathbb{R}$, and $\tilde{\tau} \in C^\infty(\mathbb{R})$.

$h \in C^\infty(\mathbb{R})$ since $g \in C^\infty(\mathbb{R})$.

It ought to be clear that the function $h(x)$ equals 1 on the closed interval $[-1, 1]$ and vanishes outside the open interval $(-2, 2)$. In particular,

$$h^{(n)}(0) = 0, \quad \forall n \in \mathbb{N}.$$

For $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, let

$$u_k(x) := x^k h(x).$$

Then it is clear that, $\forall k \in \mathbb{N} \cup \{0\}$, we have

- $u_k(x) = x^k, \forall x \in [-1, 1]$, and $u_k(x) = 0, \forall x \in \mathbb{R} \setminus (-2, 2)$;
- $u_k(x) \in C^\infty(\mathbb{R})$;
-

$$u_k^{(n)}(0) = \begin{cases} n!, & n = k, \\ 0, & n \neq k, \end{cases} \quad \forall n \in \mathbb{N} \cup \{0\}, \quad (14)$$

where $u_k^{(n)}(0)$ is the n -th derivative of u_k at $x = 0$.

$\forall n \in \mathbb{N} \cup \{0\}$ the function $u_k^{(n)}(x)$ is bounded on $\mathbb{R}$, that is

$$\forall k, n \in \mathbb{N} \cup \{0\} \quad \exists L_{k,n} > 0 \quad \forall x \in \mathbb{R} \quad \rightarrow \quad |u_k^{(n)}(x)| \leq L_{k,n}. \quad (15)$$

Indeed, $\forall n \in \mathbb{N} \cup \{0\}$ we have $u_k^{(n)}(x) = 0, \forall x \in \mathbb{R} \setminus (-2, 2)$, thus it suffices to prove that $u_k^{(n)}(x)$ is bounded on $[-2, 2]$. Since $u_k^{(n)}$ is continuous on $[-2, 2]$, the claim follows from the fact that a continuous on a closed interval function attains a maximum and a minimum on the interval (see Calculus-1).

Let

$$M_k = \max\{1, |\alpha_k|, L_{k,0}, L_{k,1}, \dots, L_{k,k}\}, \quad \forall k \in \mathbb{N} \cup \{0\}, \quad (16)$$

where α_k is the k -th element of the sequence $\{\alpha_k\}_{k=0}^\infty$ and constants $L_{k,i}$ are defined in [\(15\)](#).

For $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, let

$$f_k(x) = \frac{\alpha_k}{(M_k)^k k!} u_k(M_k x).$$

Then we have

$$f_k^{(n)}(x) = \frac{\alpha_k}{(M_k)^{k-n} k!} u_k^{(n)}(M_k x), \quad \forall k, n \in \mathbb{N} \cup \{0\}, \quad \forall x \in \mathbb{R}, \quad (17)$$

and, due to [Equality \(14\)](#),

$$f_k^{(n)}(0) = \begin{cases} \alpha_n, & n = k, \\ 0, & n \neq k, \end{cases} \quad \forall k, n \in \mathbb{N} \cup \{0\}.$$

Note that, $\forall n \in \mathbb{N} \cup \{0\}$, $\forall x \in \mathbb{R}$, we have

$$\begin{aligned} \left| \sum_{k=0}^{\infty} f_k^{(n)}(x) \right| &\leq \sum_{k=0}^{\infty} |f_k^{(n)}(x)| \leq \sum_{k=0}^{\infty} \frac{|\alpha_k|}{M_k^{k-n} k!} L_{k,n} \\ &= \sum_{k=0}^{n+1} \frac{|\alpha_k|}{M_k^{k-n} k!} L_{k,n} + \sum_{k=n+2}^{\infty} \frac{1}{k!} \cdot \frac{|\alpha_k|}{M_k} \cdot \frac{L_{k,n}}{M_k} \cdot \frac{1}{M_k^{k-n-2}} \\ &\leq C_n + \sum_{k=n+2}^{\infty} \frac{1}{k!}, \end{aligned} \quad (18)$$

where $C_n \in \mathbb{R}$. Here we used $|\alpha_k|/M_k \in [0, 1]$, $L_{k,n}/M_k \in [0, 1]$ for $k > n$, and $1/M_k^{k-n-2} \in (0, 1]$ for $k \geq n+2$, due to [\(16\)](#).

Since the series $\sum_{k=n+2}^{\infty} \frac{1}{k!}$ converges (due to [Claim 2.11](#)), due to [Inequality \(18\)](#) and [Claim 5.7](#) [The Weierstrass M-test], $\forall n \in \mathbb{N} \cup \{0\}$ the functional series $\sum_{k=0}^{\infty} f_k^{(n)}(x)$ converges uniformly on $\mathbb{R}$. Therefore, due to [Claim 6.13](#), we have

$$f^{(n)}(x) = \sum_{k=0}^{\infty} f_k^{(n)}(x), \quad \forall n \in \mathbb{N}, \forall x \in \mathbb{R}, \quad (19)$$

where we defined

$$f(x) := \sum_{k=0}^{\infty} f_k(x), \quad \forall x \in \mathbb{R}.$$

It remains to note that:

- $f \in C^\infty(\mathbb{R})$ due to [\(19\)](#);
-

$$f^{(n)}(0) = \sum_{k=0}^{\infty} f_k^{(n)}(0) = \alpha_n, \quad \forall n \in \mathbb{N} \cup \{0\}.$$

□

Claim 8.7 (Pathological Examples of Smooth Functions). There exists a smooth function f on $\mathbb{R}$ such that

1. its Taylor series $T_{f,0}(x)$ has zero radius of convergence;
2. it is not analytic at any point.

Proof. 1) Let $a_0 = 1$, $a_k = k! k^k$, $\forall k \in \mathbb{N}$. Then, due to [Claim 8.6](#), there is $f \in C^\infty(\mathbb{R})$ such that

$$f^{(k)}(0) = a_k, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Then, due to [Def 8.2](#), we have

$$T_{f,0}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (x-0)^k = \sum_{k=0}^{\infty} k^k x^k.$$

It is clear that the radius of convergence of the series $\sum_{k=0}^{\infty} k^k x^k$ is equal to zero (see [Claim 7.2](#)).

2) Let

$$f(x) = \sum_{k=1}^{\infty} \cos(2^k x) e^{-\sqrt{2^k}}, \quad \forall x \in \mathbb{R}.$$

Then it can be shown that $f \in C^\infty(\mathbb{R})$ (this is not difficult and we leave it as an exercise) and f is nowhere analytic on $\mathbb{R}$ (the proof is difficult, so we shall merely state the fact). $\square$